

Exercise I. [course]

[15 marks]

1. Let $f, g \in \mathcal{R}_{loc}([a, b])$, $b \in \mathbb{R}$ be such that $0 \leq f(t) \leq g(t)$ for all $t \in [a, b]$. Show that
- (a) If $\int_a^b g(t) dt$ converges, then $\int_a^b f(t) dt$ converges and $0 \leq \int_a^b f(t) dt \leq \int_a^b g(t) dt$.
- (b) If $\int_a^b f(t) dt$ diverges, then $\int_a^b g(t) dt$ diverges.
2. Let $f \in \mathcal{R}_{loc}([a, b])$, $b \in \mathbb{R}$. Show that if $\int_a^b f(t) dt$ is convergent, then $\lim_{x \rightarrow b^-} \int_x^b f(t) dt = 0$.

Exercise II.

Study the nature of the following improper integrals :

[16 marks]

1. $\int_0^{+\infty} \sin(2x) dx$, 3. $\int_0^{+\infty} \frac{\sqrt{x} + x}{\sqrt{x} + x^2} dx$, 5. $\int_0^1 \frac{1}{e^x - \cos x} dx$, 7. $\int_2^{+\infty} \frac{\cos x}{\ln x} dx$,
2. $\int_0^1 \frac{e^{-x}}{\sqrt{x}} dx$, 4. $\int_0^{+\infty} \frac{\ln x}{x + e^x} dx$, 6. $\int_0^{+\infty} \frac{\cos x \sin x}{\sqrt{e^x - 1}} dx$, 8. $\int_1^2 \frac{\ln(x-1)}{\sqrt{2-x}} dx$.

Exercise III.

1. Study the nature of the following series.

[18 marks]

- (a) $\sum_{n=1}^{\infty} e^{-n^2}$, (c) $\sum_{n=1}^{\infty} \frac{\sin n}{n^3 e^n}$, (e) $\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n}$,
(b) $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n}\right)$, (d) $\sum_{n=1}^{\infty} \frac{e^{\cos(n^2)}}{\sqrt{n}}$, (f) $\sum_{n=2}^{\infty} \ln\left(1 + \frac{(-1)^n}{n^2}\right)$.

2. Let $\alpha \in \mathbb{R}$. Study according to the values of α the nature of $\sum_{n=1}^{\infty} \frac{\arctan n^{2\alpha}}{n^\alpha}$.

(Hint : Discuss three cases, $\alpha > 0$, $\alpha = 0$ and $\alpha < 0$, and you may use $\arctan x \sim_0 x$.)

Exercise IV.

Calculate the following limits :

[6 marks]

- a. $\lim_{x \rightarrow 0^+} \int_x^{2x} \frac{\sqrt{\sin t}}{t\sqrt{t}} dt$, b. $\lim_{x \rightarrow +\infty} \int_x^{2x} \frac{\cos t}{t} dt$.

Exercise V.

Let $u_n \geq 0$ and $v_n \geq 0$ for all $n \in \mathbb{N}^*$. Assume that $\sum_{n=1}^{\infty} u_n$ and $\sum_{n=1}^{\infty} v_n$ are convergent.

[6 marks]

1. Show that the series $\sum_{n=1}^{\infty} (u_n)^2$ is convergent.
2. Prove that the series $\sum_{n=1}^{\infty} (u_n v_n)$ is convergent. (Hint : $ab \leq \frac{1}{2}(a^2 + b^2)$.)
3. Is part 1. remains true if the sequence (u_n) does not have constant sign? Justify your answer.

Exercise VI.

Let $0 < a < 1$ and $I(a) = \int_0^1 \frac{dx}{x^{\frac{1}{4}}(1+ax)}$, $J(a) = \int_0^a \frac{dx}{x^{\frac{1}{4}}(1+ax)}$, $K(a) = \int_a^1 \frac{dx}{x^{\frac{1}{4}}(1+ax)}$.

[9 marks]

1. Using the first Mean value theorem, show that $\lim_{a \rightarrow 0} K(a) = \frac{4}{3}$.
2. Show that the improper integral $I(a)$ is convergent. Deduce the value of $\lim_{a \rightarrow 0} J(a)$.
3. Deduce the value of $\lim_{a \rightarrow 0} I(a)$.

Exercise 1 1. Given: $f, g \in R_{loc}([a, b))$, $b \in \mathbb{R}$ and $0 \leq f(t) \leq g(t) \forall t \in [a, b)$.

(a) $\int_a^b g(t) dt$ conv $\stackrel{(3)}{\Rightarrow} G(x) = \int_a^x g(t) dt$ is bdd $\forall x \in [a, b)$

$\stackrel{(3)}{\Rightarrow} F(x) = \int_a^x f(t) dt$ is bdd
and $0 \leq F(x) \leq \int_a^x f(t) dt \leq \int_a^x g(t) dt = G(x)$

$\stackrel{(1)}{\Rightarrow} \int_a^b f(t) dt$ is conv ($\because f \geq 0$)

Now, since $0 \leq \int_a^x f(t) dt \leq \int_a^x g(t) dt \quad \forall x \in [a, b)$,

then by passing to the limit as $x \rightarrow b^-$, we obtain

$\stackrel{(2)}{\Rightarrow} 0 \leq \int_a^b f(t) dt \leq \int_a^b g(t) dt.$

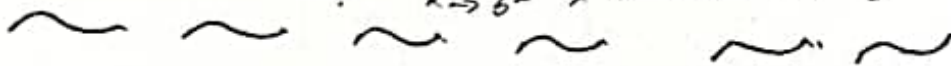
(b) Assume $\int_a^b g(t) dt$ conv, then by (a) $\int_a^b f(t) dt$ conv. $\checkmark$

2. Given $f \in R_{loc}([a, b))$, $b \in \mathbb{R}$.

$\int_a^b f(t) dt$ conv $\stackrel{(1)}{\Rightarrow} \int_a^b f(t) dt = \int_a^x f(t) dt + \int_x^b f(t) dt \quad \forall x \in [a, b)$

$\stackrel{(2)}{\Rightarrow} \lim_{x \rightarrow b^-} \int_a^b f(t) dt = \lim_{x \rightarrow b^-} \int_a^x f(t) dt + \lim_{x \rightarrow b^-} \int_x^b f(t) dt$

$\stackrel{(1)}{\Rightarrow} \lim_{x \rightarrow b^-} \int_x^b f(t) dt = 0$



Exercise II Study the Nature of the following imp integrals.

1. $\int_0^{+\infty} \sin(2x) dx = \lim_{x \rightarrow +\infty} \int_0^x \sin 2x dx = \lim_{x \rightarrow +\infty} \frac{1}{2} [\cos 2x - \cos 0] \nexists$ (div)

2. $\int_0^1 \frac{e^{-x}}{\sqrt{x}} dx$; $\frac{e^{-x}}{\sqrt{x}} \sim \frac{1}{x^{1/2}}$ (positive near 0+) (conv) by Riemann & eqn

3. $\int_0^{+\infty} \frac{\sqrt{x} + x}{\sqrt{x} + x^2} dx$; $\frac{\sqrt{x} + x}{\sqrt{x} + x^2} \sim \frac{x}{x^2} = \frac{1}{x}$ (pos); (div) by Riemann & eq. test

4. $\int_0^{+\infty} \frac{\ln x}{x + e^x} dx$; near 0+: $\frac{\ln x}{x + e^x} \sim \frac{\ln x}{e^x} = \frac{1}{\ln x}$ (conv) Bertrand

near + ∞ : $x^{\alpha} \frac{\ln x}{x + e^x} \rightarrow 0$ ($\alpha = 100$) (conv)

Hence the imp int 4. is conv.

5. $\int_0^1 \frac{1}{e^x - \cos x} dx$; $\frac{1}{e^x - \cos x} \sim \frac{1}{(1+x) - 1} = \frac{1}{x}$ (pos) (div) Riemann & eq. test

6. $\int_0^{+\infty} \frac{\cos x \sin x}{\sqrt{e^x - 1}} dx$; near 0+: $\frac{\cos x \sin x}{\sqrt{e^x - 1}} \sim \frac{x}{\sqrt{x}} = \frac{1}{x^{1/2}}$ (conv) Riemann & eq. test

near + ∞ : $x^{\alpha} \frac{\cos x \sin x}{\sqrt{e^x - 1}} \rightarrow 0$ ($\alpha = 100$) (conv)

or you may use Abel-Dirichlet test.

7. $\int_2^{+\infty} \frac{\cos x}{\ln x} dx$; $g(x) = \frac{1}{\ln x}$ dec & $g(x) \rightarrow 0$ as $x \rightarrow +\infty$ / Abel $\int_2^{+\infty} \frac{\cos x}{\ln x} dx$ (conv)
 $|F(x)| = \left| \int_2^x \cos x dx \right| \leq 2$ / Dirichlet

8. $\int_1^2 \frac{\ln(x-1)}{\sqrt{2-x}} dx$; near 1+: $\frac{\ln(x-1)}{\sqrt{2-x}} \sim \ln(x-1)$ (neg)

$\int_1^{1.5} \ln(x-1) dx = \int_0^{0.5} \ln u du$ (conv) Bertrand

near 2-: $\frac{\ln(x-1)}{\sqrt{2-x}} = \frac{\ln(1 + \frac{x-2}{1})}{\sqrt{2-x}} \sim \frac{1}{2} \frac{1}{(2-x)^{1/2}}$ (conv)

Hence the imp int 8. is conv.

Exercise III 1. Study the nature of the following series.

(a) $\sum_{n=1}^{+\infty} e^{-n^2}$; $n^2 \cdot e^{-n^2} \xrightarrow{n \rightarrow +\infty} 0$; $\alpha = 2 > 1$ (conv)

(b) $\sum_{n=1}^{+\infty} \cos(1/n)$; $\lim_{n \rightarrow +\infty} \cos(1/n) = 1 \neq 0$ (div)

(c) $\sum_{n=1}^{+\infty} \frac{\sin n}{n^3 e^n}$; $n^3 e^n \frac{\sin n}{n^3 e^n} \rightarrow 0$; $\alpha = (e > 1)$ (conv)

(d) $\sum_{n=1}^{+\infty} \frac{e^{\cos(n^2)}}{\sqrt{n}}$; $\cos(n^2) \geq -1 \Rightarrow e^{\cos(n^2)} \geq e^{-1} \Rightarrow \frac{e^{\cos(n^2)}}{\sqrt{n}} \geq e^{-1} \cdot \frac{1}{\sqrt{n}} > 0$
 Since $\sum \frac{1}{\sqrt{n}}$ div (Riemann), then the series (div) by comp
 or $n^{3/4} u_n \xrightarrow{n \rightarrow +\infty} +\infty$ & $\alpha = 3/4 < 1$ (div).

(e) $\sum_{n=2}^{+\infty} \frac{(-1)^n}{n \ln n}$; $b_n = \frac{1}{n \ln n} > 0$, dec & $\xrightarrow{n \rightarrow +\infty} 0$, (conv) Alternating Series test

(f) $\sum_{n=2}^{+\infty} \ln(1 + \frac{(-1)^n}{n^2})$; $|\ln(1 + \frac{(-1)^n}{n^2})| \sim \left| \frac{(-1)^n}{n^2} \right| = \frac{1}{n^2}$ (absol conv hence conv)

or $\ln(1 + \frac{(-1)^n}{n^2}) = \frac{(-1)^n}{n^2} + o(\frac{(-1)^n}{n^2})$; $\sum \frac{(-1)^n}{n^2}$ conv absol $\Rightarrow \sum o(\frac{(-1)^n}{n^2})$ conv absol.

2. Study according to the value of $\alpha \in \mathbb{R}$ the nature of $\sum_{n=1}^{+\infty} \frac{\arctan n^{2\alpha}}{n^\alpha}$

(i) $\alpha > 0$: $\frac{\arctan n^{2\alpha}}{n^\alpha} \sim \frac{\pi/2}{n^\alpha}$ (pos) conv if $\alpha \in]1, +\infty[$

(ii) $\alpha = 0$; $\frac{\arctan n^{2\alpha}}{n^\alpha} = \frac{\arctan 1}{1} \xrightarrow{n \rightarrow +\infty} \arctan 1 \neq 0$ (div)

(iii) $\alpha < 0$: $\frac{\arctan n^{2\alpha}}{n^\alpha} \xrightarrow{n \rightarrow +\infty} \frac{n^{2\alpha}}{n^\alpha} = \frac{1}{n^{-\alpha}}$ (pos) conv if $\alpha \in]-\infty, -1[$

Thus the series conv if $\alpha \in]-\infty, -1[\cup]1, +\infty[$ (i.e. $| \alpha | > 1$)



Exercise IV Calculate the following limits.

(a) $\lim_{x \rightarrow +\infty} \int_x^{2x} \frac{\sqrt{\sin t}}{\sqrt{t}} \cdot \frac{1}{t} dt$ $\left| \begin{array}{l} f(t) = \frac{\sqrt{\sin t}}{\sqrt{t}} \text{ cont on } [x, 2x] \\ g(t) = \frac{1}{t} \text{ cont, hence limit on } [x, 2x] \end{array} \right.$

$\stackrel{\text{F.M.V.T}}{=} \lim_{x \rightarrow +\infty} \frac{\sqrt{\sin c_x}}{\sqrt{c_x}} \int_x^{2x} \frac{1}{t} dt$

$\stackrel{(1)}{=} \lim_{x \rightarrow +\infty} \frac{1}{\sqrt{c_x}} \cdot \ln 2$ $\stackrel{(1)}{=} \ln 2$

By F.M.V.T $\exists x \leq c_x \leq 2x$
 $\Rightarrow x \rightarrow +\infty, c_x \rightarrow +\infty$

(b) $\lim_{x \rightarrow +\infty} \int_x^{2x} \frac{\cos t}{t} dt = \lim_{x \rightarrow +\infty} \int_x^{2x} \frac{1}{t} \frac{\cos t}{1} dt$

$= \lim_{x \rightarrow +\infty} \left[\frac{1}{t} \sin t \Big|_x^{2x} - \int_x^{2x} -\frac{1}{t^2} \sin t \right]$

$= \lim_{x \rightarrow +\infty} \left[\frac{1}{2x} \sin 2x - \frac{1}{x} \sin x + \int_x^{2x} \frac{1}{t^2} \sin t \right]$

$= \lim_{x \rightarrow +\infty} \frac{\sin 2x}{2x} - \lim_{x \rightarrow +\infty} \frac{\sin x}{x} + \lim_{x \rightarrow +\infty} \int_x^{2x} \frac{1}{t^2} \sin t$

$= \lim_{x \rightarrow +\infty} \frac{\sin c_x}{2x} - \lim_{x \rightarrow +\infty} \frac{\sin c_x}{x} + \lim_{x \rightarrow +\infty} \int_x^{2x} \frac{1}{t^2} \sin t$

$\stackrel{(1)}{=} \lim_{x \rightarrow +\infty} \frac{\sin c_x}{2x} = 0$ $\left(\begin{array}{l} x \leq c_x \leq 2x \\ c_x \rightarrow +\infty \text{ as } x \rightarrow +\infty \end{array} \right)$

Exercise V Given: $u_n > 0, v_n > 0, \sum u_n$ & $\sum v_n$ are conv.

1. $\sum u_n$ conv $\Rightarrow u_n \rightarrow 0 \Rightarrow (u_n)$ bdd $\Rightarrow u_n^2 \leq M u_n$ $\stackrel{(1)}{\Rightarrow} \sum u_n^2$ conv. (for some $M > 0$)

Hence By comp test $\sum u_n^2$ is conv.

2. $u_n v_n \leq \frac{1}{2} u_n^2 + \frac{1}{2} v_n^2$. Since $\sum u_n, \sum v_n$ are conv, then $\sum u_n^2$ & $\sum v_n^2$ are conv by 1. Consequently, by comparison test $\sum u_n v_n$ is conv. (since $0 \leq u_n v_n$)

3. No. $\stackrel{(1)}{\Rightarrow}$

Take $u_n = \frac{1}{n^{1/2}}$, $\sum u_n$ conv but $\sum u_n^2 = \sum \frac{1}{n}$ div.

Exercise VI

1. $\lim_{a \rightarrow 0} K(a) = \lim_{a \rightarrow 0} \int_a^1 \frac{1}{x^{1/4}(1+qx)} dx = \lim_{a \rightarrow 0} \frac{1}{1+ac} \int_a^1 \frac{1}{x^{1/4}} dx$ $\left| \begin{array}{l} a \leq c \leq 1; c \rightarrow 1 \\ f(x) = \frac{1}{x^{1/4}} \text{ cont on } [a, 1] \end{array} \right.$

$\stackrel{(1)}{=} \lim_{a \rightarrow 0} \frac{1}{1+ac} \cdot \frac{4}{3} [1 - a^{3/4}] = \frac{4}{3}$

2. $\frac{1}{x^{1/4}(1+qx)} \sim \frac{1}{x^{1/4}}$ (conv) $\stackrel{(1)}{\Rightarrow} \lim_{a \rightarrow 0} J(a) = 0$ (By Ex I, 2.) or by Cauchy crit

3. $I(a) = J(a) + K(a) \Rightarrow \lim_{a \rightarrow 0} I(a) = 0 + \frac{4}{3} = \frac{4}{3}$